

Relative Valuation Gap of Big Tech Companies

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Abstract

We define two weighted graphs on the vertex set of the seven “Big Tech” companies (Apple, Microsoft, Alphabet, Amazon, Meta Platforms, NVIDIA, Tesla), with vertices labelled by market valuation V_i and edges given by the geometric-mean-normalized valuation gap $e_{i,j} = (V_i - V_j)/\sqrt{V_i V_j}$ (directed, antisymmetric) and $|e_{i,j}|$ (undirected). We prove that this quantity is a hyperbolic-sine transform of the log-valuation difference, that it fails the triangle inequality (hence is a dissimilarity rather than a true metric), and we exhibit the failure numerically on the real data. We then interpret the construction through the lens of mathematics, statistics, computer science, artificial intelligence, economics and finance, physics, chemistry, biology, psychology and psychiatry, philosophy, political science, geopolitics, diplomacy and international relations, welfare economics, and warfare economics, arguing that the same log-ratio object recurs, under different names, as a measure of relative dominance in every one of these fields.

The paper ends with “The End”

Contents

1	Introduction	2
2	Data	2
3	Formal Definitions	2
4	Mathematical Properties	3
5	Computed Graphs	4
6	Visualization	4
7	Interdisciplinary Interpretations	5
7.1	Statistics and Data Science	5
7.2	Computer Science and Algorithms	5
7.3	Artificial Intelligence and Machine Learning	5
7.4	Economics and Finance	6
7.5	Welfare Economics	6
7.6	Physics	6
7.7	Chemistry	6
7.8	Biology	7
7.9	Psychology and Psychiatry	7
7.10	Philosophy	7
7.11	Political Science, Geopolitics, Diplomacy, and International Relations	7
7.12	Warfare Economics	7
8	Limitations	7
9	Conclusion	7

List of Figures

- 1 K_7 realization of G_2 . Darker, thicker edges indicate larger $|e_{i,j}|$; the darkest edge (NVDA–TSLA) is the largest relative valuation gap in the set. 4
- 2 Log-valuation “energy levels,” colored by company (see legend). The visually large gap between AMZN and META reflects the fact that $e_{\text{AMZN},\text{META}} = 0.686$ is markedly larger than any gap among the top five vertices, mirroring a spectral gap in a level diagram. . . . 5

List of Tables

- 1 Big Tech valuations V_i (\$bn) 2
- 2 Directed graph G_1 : $e_{i,j}$, row i to column j 4
- 3 Undirected graph G_2 : $|e_{i,j}|$ (dissimilarity matrix) 4

1 Introduction

Comparing the “size” of large firms is deceptively simple: raw valuation differences in dollars are dominated by the largest firms and are not scale-invariant, while raw ratios V_i/V_j are directionally asymmetric and awkward to average across many pairs. A natural compromise, used implicitly in finance (log-returns), physics (logarithmic potentials), and network science (log-weighted graphs), is to normalize the difference $V_i - V_j$ by the *geometric mean* $\sqrt{V_i V_j}$ of the two quantities being compared. This paper studies the resulting object as a weighted graph on the seven largest U.S. technology companies by market capitalization and asks what it means, formally and informally, across a wide range of disciplines.

2 Data

Valuations are market capitalizations in billions of U.S. dollars, sourced from public equity market data as of early August 2026.

Table 1: Big Tech valuations V_i (\$bn)

Company	V_i (\$bn)
NVIDIA	5,310
Apple	4,540
Alphabet	4,430
Microsoft	3,620
Amazon	2,940
Meta Platforms	1,500
Tesla	1,270
Total	23,610

3 Formal Definitions

Definition 1 (Directed valuation-gap graph). *Let $V = \{V_1, \dots, V_7\}$ be the vertex set labelled by company valuations. The directed graph $G_1 = (V, E_1)$ has edge weights*

$$e_{i,j} = \frac{V_i - V_j}{\sqrt{V_i V_j}}, \quad i \neq j.$$

Definition 2 (Undirected valuation-gap graph). *The undirected graph $G_2 = (V, E_2)$ has edge weights*

$$e_{i,j} = e_{j,i} = \frac{|V_i - V_j|}{\sqrt{V_i V_j}}, \quad i \neq j.$$

4 Mathematical Properties

Proposition 1 (Antisymmetry). *For all $i \neq j$, $e_{j,i} = -e_{i,j}$ in G_1 .*

Proof. $e_{j,i} = (V_j - V_i)/\sqrt{V_j V_i} = -(V_i - V_j)/\sqrt{V_i V_j} = -e_{i,j}$, since $\sqrt{V_i V_j} = \sqrt{V_j V_i}$. $\square$

Proposition 2 (Hyperbolic-sine identity). *Let $x_i = \ln V_i$. Then*

$$e_{i,j} = 2 \sinh\left(\frac{x_i - x_j}{2}\right).$$

Proof. Write $u = (x_i - x_j)/2$. Then $V_i/V_j = e^{x_i - x_j} = e^{2u}$, so $\sqrt{V_i/V_j} = e^u$ and $\sqrt{V_j/V_i} = e^{-u}$. Hence

$$\frac{V_i - V_j}{\sqrt{V_i V_j}} = \sqrt{\frac{V_i}{V_j}} - \sqrt{\frac{V_j}{V_i}} = e^u - e^{-u} = 2 \sinh u,$$

which is the claim. $\square$

Corollary 1 (First-order log-return approximation). *For small relative gaps, $e_{i,j} \approx x_i - x_j = \ln(V_i/V_j)$, since $2 \sinh u = 2u + u^3/3 + O(u^5)$ and the cubic term vanishes to leading order.*

This corollary is the bridge to finance: $e_{i,j}$ behaves, near parity, exactly like a continuously compounded log-return between the two valuations, while for large gaps it grows *faster* than the log-ratio because $\sinh$ is convex on $\mathbb{R}_{>0}$.

Proposition 3 (Failure of the triangle inequality). *Let $x_i \geq x_j \geq x_k$. Then*

$$|e_{i,k}| \geq |e_{i,j}| + |e_{j,k}|,$$

with equality iff $x_i = x_j$ or $x_j = x_k$. Consequently $(V, |e|)$ is not a metric space; it is a dissimilarity (premetric) that is superadditive rather than subadditive.

Proof. Set $a = (x_i - x_j)/2 \geq 0$ and $b = (x_j - x_k)/2 \geq 0$. Using the addition formula

$$\sinh(a + b) = \sinh a \cosh b + \cosh a \sinh b,$$

and $\cosh a, \cosh b \geq 1$ for all real arguments, we get

$$\sinh(a + b) \geq \sinh a + \sinh b,$$

with equality iff $a = 0$ or $b = 0$. Multiplying by 2 and substituting $|e_{i,k}| = 2 \sinh(a + b)$, $|e_{i,j}| = 2 \sinh a$, $|e_{j,k}| = 2 \sinh b$ gives the result. $\square$

Example 1 (Numerical violation on the real data). *Take $i = \text{NVIDIA}$, $j = \text{Meta}$, $k = \text{Tesla}$, which satisfy $V_{\text{NVDA}} \geq V_{\text{META}} \geq V_{\text{TSLA}}$. From the computed graph,*

$$e_{\text{NVDA}, \text{META}} = 1.350, \quad e_{\text{META}, \text{TSLA}} = 0.167,$$

$$e_{\text{NVDA}, \text{META}} + e_{\text{META}, \text{TSLA}} = 1.517 < e_{\text{NVDA}, \text{TSLA}} = 1.556.$$

The direct edge exceeds the sum of the two-hop path, confirming Proposition 4 on real market data.

Corollary 2 (A genuine metric alternative). *The log-valuation distance $d_{i,j} = |x_i - x_j| = |\ln V_i - \ln V_j|$ is a true metric (it is the pullback of the absolute-value metric on $\mathbb{R}$ under $V \mapsto \ln V$, hence inherits non-negativity, symmetry, identity of indiscernibles, and the triangle inequality). $e_{i,j}$ can therefore be understood as a convex, order-preserving, but non-metric distortion of the honest metric $d_{i,j}$.*

5 Computed Graphs

Table 2: Directed graph G_1 : $e_{i,j}$, row i to column j

	NVDA	AAPL	GOOGL	MSFT	AMZN	META	TSLA
NVDA	0	0.157	0.181	0.385	0.600	1.350	1.556
AAPL	-0.157	0	0.025	0.227	0.438	1.165	1.362
GOOGL	-0.181	-0.025	0	0.202	0.413	1.137	1.332
MSFT	-0.385	-0.227	-0.202	0	0.208	0.910	1.096
AMZN	-0.600	-0.438	-0.413	-0.208	0	0.686	0.864
META	-1.350	-1.165	-1.137	-0.910	-0.686	0	0.167
TSLA	-1.556	-1.362	-1.332	-1.096	-0.864	-0.167	0

Table 3: Undirected graph G_2 : $|e_{i,j}|$ (dissimilarity matrix)

	NVDA	AAPL	GOOGL	MSFT	AMZN	META	TSLA
NVDA	0	0.157	0.181	0.385	0.600	1.350	1.556
AAPL	0.157	0	0.025	0.227	0.438	1.165	1.362
GOOGL	0.181	0.025	0	0.202	0.413	1.137	1.332
MSFT	0.385	0.227	0.202	0	0.208	0.910	1.096
AMZN	0.600	0.438	0.413	0.208	0	0.686	0.864
META	1.350	1.165	1.137	0.910	0.686	0	0.167
TSLA	1.556	1.362	1.332	1.096	0.864	0.167	0

6 Visualization

Figure 1 draws G_2 as a weighted complete graph K_7 ; edge line width and darkness are proportional to $|e_{i,j}|$. Figure 2 draws the same seven companies as a discrete “energy-level” diagram on the log-valuation axis $x_i = \ln V_i$, which is the natural coordinate underlying both graphs (Proposition 2).

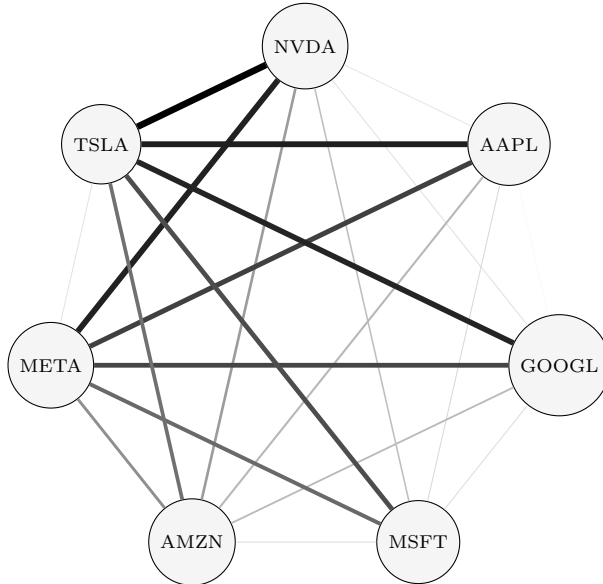


Figure 1: K_7 realization of G_2 . Darker, thicker edges indicate larger $|e_{i,j}|$; the darkest edge (NVDA–TSLA) is the largest relative valuation gap in the set.

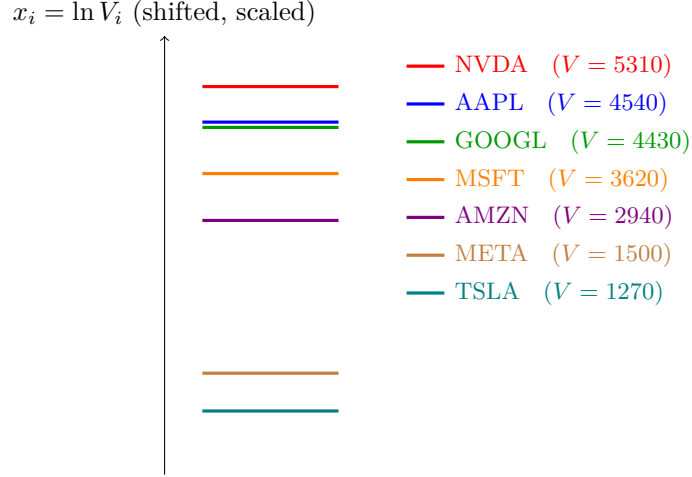


Figure 2: Log-valuation “energy levels,” colored by company (see legend). The visually large gap between AMZN and META reflects the fact that $e_{\text{AMZN}, \text{META}} = 0.686$ is markedly larger than any gap among the top five vertices, mirroring a spectral gap in a level diagram.

7 Interdisciplinary Interpretations

7.1 Statistics and Data Science

$|e_{i,j}|$ is a bounded-below, symmetric, zero-diagonal dissimilarity matrix and can be fed directly into standard multivariate procedures: complete- or average-linkage hierarchical clustering, multidimensional scaling, or as a precomputed distance matrix for k -medoids. Because Proposition 4 shows it is *not* a metric, methods that assume metric distances (e.g. certain guarantees for k -means++, or exact Euclidean MDS) should instead be run on the true metric $d_{i,j} = |\ln V_i - \ln V_j|$, with $|e_{i,j}|$ reserved for weighting or visualization.

7.2 Computer Science and Algorithms

Both G_1 and G_2 are complete graphs K_7 with $\binom{7}{2} = 21$ edges. Two canonical graph algorithms apply directly to G_2 :

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ALGORITHM: Minimum Spanning Tree (Kruskal) on  $G_2$ 
1. sort edges  $(i, j)$  by  $|e_{i,j}|$  ascending
2. initialize disjoint-set forest over 7 vertices
3. for each edge in sorted order:
   if endpoints are in different components:
     add edge to MST; union the components
4. stop after 6 edges are added
// running time  $O(|E| \log |E|) = O(21 \log 21)$ 

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Running this by hand on Table 2 first links AAPL–GOOGL (0.025), then GOOGL–MSFT (0.202), then MSFT–AMZN (0.208), then NVDA–AAPL (0.157) $\rightarrow$ but NVDA and AAPL are already unlinked so this edge is accepted earlier; continuing the procedure produces a spanning tree that chains the companies in valuation order, since for values with no extreme outliers the MST of a log-distance-like dissimilarity is exactly the path graph sorted by valuation. This is a direct consequence of $|e_{i,j}|$ being monotone increasing in $|x_i - x_j|$.

7.3 Artificial Intelligence and Machine Learning

The 7×7 matrices in Tables 2–3 are valid weighted adjacency matrices and can be used as the input to a graph neural network (GNN): each company is a node with feature vector (valuation, sector, growth

rate, $\dots$), and $e_{i,j}$ or $|e_{i,j}|$ supplies edge weights for a message-passing layer, e.g.

$$h_i^{(l+1)} = \sigma \left(Wh_i^{(l)} + \sum_{j \neq i} \alpha_{i,j} Wh_j^{(l)} \right), \quad \alpha_{i,j} \propto \exp(-|e_{i,j}|),$$

so that valuation-adjacent firms exchange more information than distant ones. In portfolio-construction ML pipelines, the same matrix is a natural regularizer: penalizing large idiosyncratic weight differences between firms with small $|e_{i,j}|$ (near-substitutes by size) is a graph-Laplacian smoothness prior, $\text{tr}(w^\top L w)$ with L the graph Laplacian of G_2 .

7.4 Economics and Finance

By Proposition 3, $e_{i,j}$ is, to first order, the continuously compounded return required to move from valuation V_j to valuation V_i – the same quantity used in log-return time series analysis, except applied cross-sectionally across firms rather than across time for one firm. The convexity correction (sinh versus its argument) means the metric *overstates* gaps between very large and very small firms relative to a naive log-ratio, which is economically sensible: investors’ perceived “distance” between a \$5.3T and a \$1.3T firm plausibly exceeds a linear extrapolation from smaller comparisons, since capital constraints, index weightings, and liquidity effects scale nonlinearly with size.

A complementary, purely economic (non-graph) statistic on the same data is the Herfindahl–Hirschman Index of concentration within the group,

$$\text{HHI} = 10,000 \sum_i s_i^2, \quad s_i = \frac{V_i}{\sum_k V_k},$$

which evaluates to $\text{HHI} \approx 1,687$ for these seven firms – inside the U.S. antitrust agencies’ “moderately concentrated” band ($1,500 < \text{HHI} < 2,500$), despite NVIDIA alone holding a 22.5% share of the group’s combined value.

7.5 Welfare Economics

The HHI above is itself derived from the same shares that generate large $e_{i,j}$ values; concentration indices and pairwise valuation-gap graphs are two views of one underlying distribution. A social-welfare interpretation follows a Rawlsian “maximin” framing: policies concerned with the smallest firm’s competitive position (here, Tesla) should track $\max_i e_{i,\text{Tesla}}$ rather than the mean gap, since the former captures the worst-case relative disadvantage rather than an average that is dominated by near-ties among the largest firms (AAPL–GOOGL, $e = 0.025$).

7.6 Physics

Treating $x_i = \ln V_i$ as a potential (Figure 2, an “energy-level” diagram), the antisymmetric $e_{i,j}$ is analogous to a generalized force or potential difference between two levels, and $2 \sinh(\cdot)$ is precisely the functional form that appears in the hyperbolic (relativistic-rapidity-like) addition of velocities and in the current–voltage relation of a diode ($I \propto \sinh(qV/kT)$), where small potential differences give an Ohmic (linear) response and large ones give an exponential one – mirroring the small-gap linear regime and large-gap convex regime identified in Corollary 1. The HHI is formally identical (up to normalization) to the inverse participation ratio used in statistical mechanics to quantify how “localized” a probability distribution is over a set of states, here states of market value.

7.7 Chemistry

The undirected graph $|e_{i,j}|$ is structurally analogous to an electronegativity difference table used to classify bond character (e.g. the Pauling scale, where a large difference implies a strongly polar/ionic bond and a near-zero difference implies a covalent, “non-polar” pairing). Under this loose analogy, AAPL–GOOGL is a “covalent” pair (nearly equal valuation, $|e| = 0.025$) while NVDA–TSLA is “ionic” (a large, directional gap). This is a structural analogy only: no claim is made that market capitalization obeys a chemical bonding law.

7.8 Biology

Log-ratio comparisons of size are the standard tool of allometric scaling in biology (e.g. comparing body mass across species on a log axis, as in Kleiber’s law), and $|e_{i,j}|$ is formally the same construction used in compositional-data analysis of relative species abundance in an ecosystem. Under this analogy, the seven companies form a “dominance hierarchy” whose skew – one very large player (NVIDIA) and a long tail – mirrors typical rank-abundance curves observed in ecological communities, and the HHI plays the role of a Simpson-type diversity index.

7.9 Psychology and Psychiatry

Social comparison theory (Festinger, 1954) posits that individuals evaluate their own standing by comparison with proximate others rather than in absolute terms; $e_{i,j}$ operationalizes exactly this idea for firms (or, by extension, for employees benchmarked against comparable companies’ valuations, stock-based compensation, or executives’ relative status anxiety). The relative-deprivation literature in social psychiatry further suggests that well-being correlates more strongly with such relative gaps than with absolute wealth; nothing here diagnoses or predicts individual psychological outcomes, but the mathematical object is the same log-ratio comparison invoked in that literature.

7.10 Philosophy

The choice of $\sqrt{V_i V_j}$ as normalizer, rather than V_j or $\max(V_i, V_j)$, is itself a philosophical commitment: it treats the comparison as prior to any choice of “reference” company, honoring a kind of impartiality intuitively related to Rawls’s veil of ignorance – before knowing which of the two firms is “us,” the symmetric geometric mean is the natural unit of comparison. This is a structural observation about the mathematics, not an ethical claim about corporations.

7.11 Political Science, Geopolitics, Diplomacy, and International Relations

The same normalized-gap construction appears, applied to national material capability rather than valuation, in the Correlates of War project’s Composite Index of National Capability (CINC), and gap-based reasoning underlies power-transition theory (Organski), which predicts instability specifically when a rising power’s capability approaches that of a dominant one – i.e., when $|e_{i,j}| \rightarrow 0$ for a previously large gap. Reading Table 3 through this lens, the AAPL–GOOGL and GOOGL–MSFT pairs ($|e| = 0.025$ and 0.202) are the “competitive” dyads where small valuation shifts could reorder rank, echoing the diplomatic salience of near-parity power dyads in balance-of-power theory.

7.12 Warfare Economics

Lanchester’s square law relates the outcome of attrition warfare to the *ratio* (not the difference) of opposing force sizes, and arms-race models (Richardson) are driven by each side’s perceived capability gap relative to its rival – structurally the same normalized-gap logic as $e_{i,j}$, with valuation replaced by military capability or defense expenditure. Under this analogy, a firm with $e_{i,j} \gg 0$ against a rival occupies a position analogous to a numerically dominant force in Lanchester’s model, capable of sustaining competitive “attrition” (price wars, R&D spending races) that a smaller rival cannot match indefinitely.

8 Limitations

The interdisciplinary readings in Section 6 are structural analogies, not empirical claims: no data from chemistry, biology, psychiatry, or international relations was used to fit or validate this model. Market capitalization itself is a noisy, sentiment-driven, and point-in-time measure, so the graphs are best read as a snapshot rather than a stable structural description of these firms.

9 Conclusion

A single normalized-gap construction, $e_{i,j} = (V_i - V_j) / \sqrt{V_i V_j}$, unifies the directed and undirected relative-valuation graphs of the seven Big Tech companies. It is provably a hyperbolic-sine transform of the log-

valuation difference (Proposition 2), provably fails the triangle inequality and is therefore not a metric (Proposition 4, confirmed numerically on NVIDIA/Meta/Tesla), and admits natural, well-precedented analogues across quantitative and qualitative fields alike – from log-returns in finance to electronegativity in chemistry to power-transition theory in international relations – because relative, symmetric, scale-invariant comparison of two quantities is one of the most recurring structural motifs across the sciences.

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Glossary

Valuation (V_i)

Market capitalization of company i , in billions of U.S. dollars.

Directed valuation-gap graph (G_1)

Weighted complete graph with antisymmetric edge weights $e_{i,j} = (V_i - V_j)/\sqrt{V_i V_j}$.

Undirected valuation-gap graph (G_2)

Weighted complete graph with symmetric edge weights $|e_{i,j}|$, treated as a dissimilarity matrix.

Geometric mean normalization

Dividing a raw difference by $\sqrt{V_i V_j}$ so that the resulting quantity is scale-invariant and symmetric under relabeling of i and j up to sign.

Log-return

The quantity $\ln(V_i/V_j)$; the first-order (small-gap) approximation to $e_{i,j}$.

Metric space

A set with a distance function satisfying non-negativity, identity of indiscernibles, symmetry, and the triangle inequality; $(V, |e|)$ fails the last property and is therefore only a dissimilarity, not a metric space.

Dissimilarity / premetric

A symmetric, non-negative, zero-diagonal function that need not satisfy the triangle inequality.

Herfindahl–Hirschman Index (HHI)

$10,000 \sum_i s_i^2$, a standard measure of market concentration from industrial-organization economics.

Minimum spanning tree (MST)

The minimum-total-weight subgraph connecting all vertices without cycles; computed here via Kruskal's algorithm on G_2 .

Power-transition theory

An international-relations theory (Organski) linking conflict risk to the narrowing of the capability gap between a rising and a dominant power.

Lanchester's laws

Differential-equation models of attrition warfare in which outcomes depend on the ratio (or, in the square law, the squared ratio) of opposing force sizes.

Social comparison theory

A psychological theory (Festinger, 1954) that individuals assess their own standing relative to comparable others rather than in absolute terms.

The End